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Research Interests

Macroeconomics, Heterogeneous Agents, Heterogeneous Firms, Optimal Fiscal and Monetary Policy

Employment

Postdoctoral Fellow	University of Southern California	Jan. 2025 -
Ph.D. Trainee	Bank of England	Jul. 2023 - Dec. 2023
Visiting Short-term Scholar	Michigan State University	Sep. 2017 - Feb. 2018

Education

PhD Economics	Sciences Po, Paris	Aug. 2021 - Nov. 2024
MA Economics	University of Minnesota	Aug. 2018 - Aug. 2020
MSc Economics	Sao Paulo, School of Economics, FGV	Jan. 2016 - Jul. 2018
BSc Economics	University of Sao Paulo, USP	Feb. 2011 - Dec. 2015

Publications

Financial constraints and collateral crises, *Review of Economic Dynamics* (October, 2020)
Luis Araujo[1], Bernardo Guimares[2], Diego Rodrigues

Assessing the fundamental value of a wide range of asset-backed securities is costly. As a result, these assets can become information insensitive, which allows them to be used as collateral in credit transactions. In this paper, we show that while it is true that information-insensitive assets can play a liquidity role, the fact that they play this role reinforces their information insensitivity. This implies that the availability of alternative ways of financing can harm the liquidity role of assets, even if these alternatives are costly and not used in equilibrium. The reason is that such options raise the asset's sensitivity to information by increasing the relative importance of their fundamental value vis-a-vis their role as collateral.

[1] Michigan State University.
[2] Sao Paulo School of Economics, FGV.

Job Market Paper

Inventories matter for the transmission of monetary policy: uncovering the cost-of-carry channel
Diego Rodrigues, Tim Willems[3]

By setting interest rates, monetary policy affects the cost of carrying inventories. We build a model to capture the resulting "cost-of-carry channel" of monetary policy, finding it to yield interesting non-linearities. We begin with a static model showing that higher inventory costs drive firms, especially those with larger inventories, to reduce prices. Extending this to a dynamic model where firms face distinct demand shocks, we first show in a tractable model with only high and low demand shocks that firms with inventories reduce prices as carrying costs rise, while others may increase them. In aggregate, higher inventory levels make prices more responsive to tighter monetary policy. Finally, a quantitative model with firm heterogeneity and idiosyncratic demand shocks reveals that prices should be more sensitive to the stance of monetary policy when inventory levels are higher (effectively leaving sellers with less market power). By drawing on data from the US housing market, we are able to test this hypothesis – finding strong support for the cost-of-carry channel. Central banks may therefore wish to pay close attention to inventory levels, as they could matter for the strength of monetary policy transmission to inflation.

Presented: Sciences Po Seminars and Macroeconomics Reading Group.

Working Papers

The Welfare of Nations: Social Preferences and the Macroeconomy
François Le Grand[4], Xavier Ragot[5], Diego Rodrigues

We present a Bewley-type theory of the Social Welfare Function (SWF). Agents have heterogeneous perceptions of how the planner should value the welfare of other agents, which results in so-called Individual Welfare Functions (IWFs), shaped by their life experiences. The SWF is constructed as an aggregation of IWFs, weighted by the political weights of each group of agents. We then propose an estimation strategy to compute the SWF and IWFs using the inverse optimal approach in a heterogeneous-agent model. We apply our methodology to France and the United States. France's SWF places greater emphasis on individuals with lower incomes, contrasting with the United States' SWF, which assigns greater weight to individuals with higher incomes. Finally, we simulate the fiscal system of the United States under the assumption of adopting the French SWF. Our findings indicate that the SWF significantly shapes the optimal steady-state fiscal system and equilibrium inequality.

Presented: AFSE 2022, EEA/ESEM 2022, NBER Summer Institute 2024.

Terms of trade shocks, import dependency and debt distress
Naomi Cohen[6], Venance Riblier[7], Diego Rodrigues

This paper examines the relationship between terms-of-trade shocks and sovereign debt defaults in import-dependent developing economies. While existing literature highlights a negative relationship between terms-of-trade shocks and defaults for commodity exporters, we find the opposite for commodity importers, where rising import prices can worsen debt distress and increase default risks. We incorporate trade into a sovereign default model to explore this interaction, finding that countries with a high import share are at risk of default when facing rising import prices and deteriorating terms of trade. Additionally, heavily indebted nations

[3]Bank of England and Centre for Macroeconomics.
[4]Rennes School of Business and ETH Zurich.
[5]SciencesPo, OFCE, and CNRS.
[6]Sciences Po, Paris.
[7]UC Berkeley.

risk default even with modest price increases. Our empirical analysis focuses on food as a key terms-of-trade shock, given its limited substitutability with domestic production. Using unexpected harvest shocks as an instrument to isolate food price fluctuations, we focus on Ghana, which defaulted on its external debt in 2022. We find that unexpected food price increases drive up import costs, inflation, trade imbalances, and debt. These results underscore the importance of consumption composition in assessing trade shock impacts.

Presented: 9th IMAC Workshop, Bordeaux School of Economics, 2024 ADRES Job Market Conference, RIEF Network 2024, DebtCon7 Emerging Scholars Session.

Loving or Fighting Inflation: Managing debt dynamics
Diego Rodrigues

The economic landscape post-COVID-19 and the energy crisis has underscored the critical role of government intervention in cushioning against shocks and aiding economic recovery. Focusing on the United Kingdom, we analyze optimal fiscal and monetary policies in scenarios marked by differing debt-to-GDP ratios, notably comparing the low ratio of 2007 to the high ratio of 2021. By exploring dynamic responses to shocks and the utilization of policy tools like taxation, public debt, and inflation, we offer insights into effective strategies for navigating economic uncertainties. Specifically, we highlight how higher debt-to-GDP ratios necessitate nuanced approaches to managing public debt in response to shocks. Additionally, we find that in the absence of fiscal tools, inflation can serve as an effective adjustment mechanism, suggesting that accepting moderate inflation may be optimal in certain scenarios. We use an extended Lagrangian approach for analytical results and a truncated representation of incomplete markets model for quantitative findings, offering a manageable framework for studying policy dynamics in response to economic shocks.

Presented: AEA 2024.

Work in Progress

Circulation of debts issued by the bank and the probability of bank runs

Optimal Monetary and Fiscal Policy in a World of supply shocks

A HANK model fo France

Teaching Experience

Fall 2022 **Lecturer, Macroeconomics I, Ph.D. in Economics, Sciences Po**.
Topics: Economic Growth Models, Labor Market Dynamics, Dynamic Optimization.
Github page: `https://github.com/Diego-de-Sousa-Rodrigues/Macroeconomics_I`

Fall 2022 **Teaching Assistant, Macroeconomics III, Ph.D. in Economics, Sciences Po**.
Topics: Market Structures, Dynamic Systems and Optimization, Heterogeneous Agent Models.
Github page: `https://github.com/Diego-de-Sousa-Rodrigues/Macroeconomics_III`

Spring 2022 **Lecturer, Programming Course, Ph.D. in Economics, Sciences Po**.
Topics: Terminal, GitHub, Basics in R, Advanced programming in Julia.
Github page: `https://github.com/Diego-de-Sousa-Rodrigues/IP2022`

Fall 2021 **Teaching Assistant, Macroeconomics III, Ph.D. in Economics, Sciences Po**.
Topics: Basic Asset Pricing, Dynamic Programming, Inequality, Aggregate Fluctuations.

Fall 2021	**Teaching Assistant, Macroeconomics I, Ph.D. in Economics, Sciences Po**. *Topics:* Economic Growth, Labor Markets, Unemployment, Search and Matching.
Fall 2020	**Teaching Assistant, Microeconomics, BA in Economics, University of Minnesota**. *Topics:* Households, Firms, and Industries under Competitive/Monopolistic Conditions.
Summer 2020	**Teaching Assistant, Microeconomics, BA in Economics, University of Minnesota**. *Topics:* Microeconomic Behavior of Consumers, Firms, and Markets.
Spring 2017	**Teaching Assistant, Macroeconomics II, Ph.D. in Economics, FGV**. *Topics:* Basic Macroeconomics Models in Discrete Time and Monetary Models.
Spring 2017	**Teaching Assistant, International Finance, Professional M.A. in Economics, FGV**. *Topics:* Introduction to International Finance and Open Macroeconomics.

Honors and Awards

2021 - 2024	Fellowship Sciences Po
2018 - 2019	Bert and Susan Hill Gross Fellowship University of Minnesota
2017 - 2018	Scholarship to Research Internship Abroad at Michigan State University.
2016 - 2018	Scholarship from FAPESP for Master in Economics at FGV

Language and Computer Skills

Languages	Portuguese (Native), English (Fluent), French (B2), Spanish (B1)
Computer Skills	MATLAB, Julia, R, Stata, E-Views, Gretl, LaTex

Professional Experience

2012 - 2015	Consultant, Neuding Management Consultant, Sao Paulo/SP, Brazil

References

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